OXFORD AQA
INTERNATIONAL QUALIFICATIONS

The international exam board *that puts fairness first*

International AS and A-level

Physics (9630)
PH02 Electricity, waves and particles
Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL AS PHYSICS (9630) PH02 – JANUARY 2025

This paper was accessible to students but still discriminated well. The mean mark was just above 48. There was no indication of significant time constraints.

As seen in previous series, numerically-based questions were answered better than those requiring written explanations. In several questions, the terminology used by students could be improved.

QUESTION 01

Nearly 80% of students gained both marks for this de Broglie calculation. An incorrect conversion of $0.98\ \mathrm{pm}$ was the usual error that led to only 1 mark being awarded.

QUESTION 02

Almost all students knew that the area of the current–time graph would give the amount of charge transferred. The majority of students scored all three marks; however, common errors were:

- failing to spot that the current did not go to zero at 3.0 hours
- not converting the charge unit from $\mathrm{A\ h}$ to C.

QUESTION 03

03.1 This simple wave equation calculation was found to be very accessible.

03.2 This question on polarisation discriminated well. The first marking point was for linking polarisation to some change in the microammeter reading. This change could have been the insertion of the metal grille or the rotation of the grille.

Few students explained why the microammeter reading changed between the two orientations. There was evidence that many students had an incorrect ‘picket fence’ interpretation of microwave polarisation by the grille; this was condoned.

Many students identified that polarisation is only possible with transverse waves and so the microwaves were transverse in nature.

QUESTION 04

04.1 Students found little difficulty with this determination. Nearly 80% correctly read from the graph, converted the diameter to an area, and correctly rearranged the resistivity equation for the length of the wire.

04.2 This extrapolation was dealt with very well by students in a variety of ways. The most common method was to use $y = mx + c$. Students who gained only one mark often forgot to include the y-intercept after they had found the gradient.

QUESTION 05

05.1 Many students were familiar with a potential divider circuit and corresponding equations. Most deduced that the maximum resistance of $\mathbf{R}_V$ corresponded to a voltmeter reading of $3.0\ \mathrm{V}$ and gave some relevant working for the correct answer.

05.2 This question discriminated fairly over the mark range. About 45% of students identified, for one mark, that a potential difference would now occur across the internal resistance.

Unusual phrases for this idea were often seen, for example "*a pd exists between the battery*". More conventional terminology was expected, such as "*a pd exists across the internal resistance*".

A further fifth of students successfully expressed how this additional potential difference related to the emf and voltmeter reading.

QUESTION 06

06.1 Over 85% of students gained full marks for this calculation.

06.2 This question discriminated well. As with other questions, conventional terminology was expected: for instance, "*nature frequency*" was not acceptable in place of "*natural frequency*".

06.3 This question was found to be more challenging than 06.2. A general statement that the amplitude would decrease was commonly seen but was insufficient for the first marking point. There needed to be some reference to this occurring at all frequencies, or that the maximum amplitude (at resonance) would decrease.

QUESTION 07

07.1 and 07.2 Both parts of this question discriminated fairly. Answers for 07.1 were accepted in a range of formats but diagrams had to be convincing for any marks. Sketches that resembled a diffraction grating pattern were not rewarded.

Students commonly gained the first marking point for 07.2, with about a fifth also commenting on the reduced brightness.

07.3 Students had little difficultly with this maximum order diffraction grating calculation. Those who failed to score both marks often failed to give an integer answer.

QUESTION 08

08.1 Nearly 60% of students gave a sufficient description of coherent waves for both marks. The criterion of a constant phase relationship was often missing or inadequately expressed.

08.2 Very few students identified that the path difference was $\frac{\lambda}{2}$ for full marks. A general expression of $(\mathrm{n} + 0.5)\lambda$ was not accepted as it did not relate to the position stated in the question.

Students should note that ‘out of phase’ is not equivalent to ‘anti-phase’ when describing a phase difference.

One–mark answers frequently referred to destructive interference occurring. Variations of the word “destructive” were often seen. Students are advised to know the correct terminology.

08.3 About 40% of students scored both marks here by giving a relevant explanation for $0.10\ \mathrm{m}$. The most common explanation was that the spacing between adjacent minima on a stationary wave is half a wavelength.

08.4 This question proved challenging for students to answer in clear language, with only about 20% gaining any marks. Many understood that the different distances would unequally affect the amplitudes but were not accurate in their terminology. References to “loudness” or “intensity” were accepted, instead of the correct term “amplitude”, for the first marking point.

The second marking point was often not gained because there was no initial statement about superposition occurring.

QUESTION 09

09.1 Very few students gave sufficient detail to gain full credit in this question. The best answers began with the idea that energy was transferred to a mercury atom by collision with an electron. This collision excited the atom, which then emitted ultraviolet photons during de-excitation. The idea of energy transfer by collision was frequently the missing aspect to relevant answers.

09.2 and 09.3 These calculations were found to be very accessible by students. Full marks were gained by over 60% of students in 09.2 and 80% in 09.3. Common errors were not converting mW and/or rearranging equations incorrectly.

09.4 This multi-step determination was achieved with high success by nearly 70% of students. Final answers given to one significant figure were not accepted.

QUESTION 10

10.1 Almost all students scored the mark for this reversed mean average calculation.

10.2 and 10.3 The most common error in these data analysis questions was giving answers to more than two significant figures. A less frequent mistake was using 0.77 for the lowest value in the range calculation.

10.4 Students found this contextual question challenging, with barely 25% gaining any mark. The most common successful answers recognised that a parallax error may occur when student A timed the fall. The student would perceive the pod reaching the horizontal line on the wall when the pod was actually above $1.63\ \mathrm{m}$. As a result, the measured time would be larger and the calculated terminal speed would be smaller.

A less frequently seen correct response identified that the pod may not have reached terminal speed as it passed the horizontal line. Students who made this comment often failed to realise that this would lead to a larger measured time and a smaller calculated terminal speed.

QUESTION 11

11.1 and 11.2 About two-thirds of students gained full marks in both of these calculations. Those who gained only one mark in 11.1 gave the speed of light in the glass as their final answer.

Students who failed to get full marks in 11.2 often did not identify that the angle of refraction was $61°$.

11.3 Nearly 60% of students recognised that the critical angle would be $45°$ in this calculation.

11.4 Only about half of the students gained any credit in this question. About one quarter identified that total internal reflection would occur, but did not explain why in terms of the relevant angles. This was another question where correct terminology was expected but not always seen.

SECTION C

All the questions in Section C were found to be accessible.

